

# The Agulhas Plateau and the unstable behaviour of the Bouvet triple junction through Cretaceous time, 146-66 Ma



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0.00 Ma :: CR26BAAD :: Figure 1 :: 2026 March 23



Figure 1.

1. Three widely-separated submarine plateaux - the Agulhas Plateau, the NE Georgia Bank and the Maud Rise - are loosely attributed to magmatic activity at the Bouvet triple junction in the past. This poster explains in detail how this happened by investigating the behaviour of the mid-ocean ridges around the triple junction during Cretaceous time.
2. In the South Atlantic Ocean and the Indian Ocean, the mid-ocean ridges have remained (mostly) central and stable. They may be modelled simply; there have been few ridge 'jumps'.
3. Where three ridges meet at a triple junction, instability occurs more often.

83.00 Ma :: CR26BAAD :: Figure 3 :: 2026 March 23

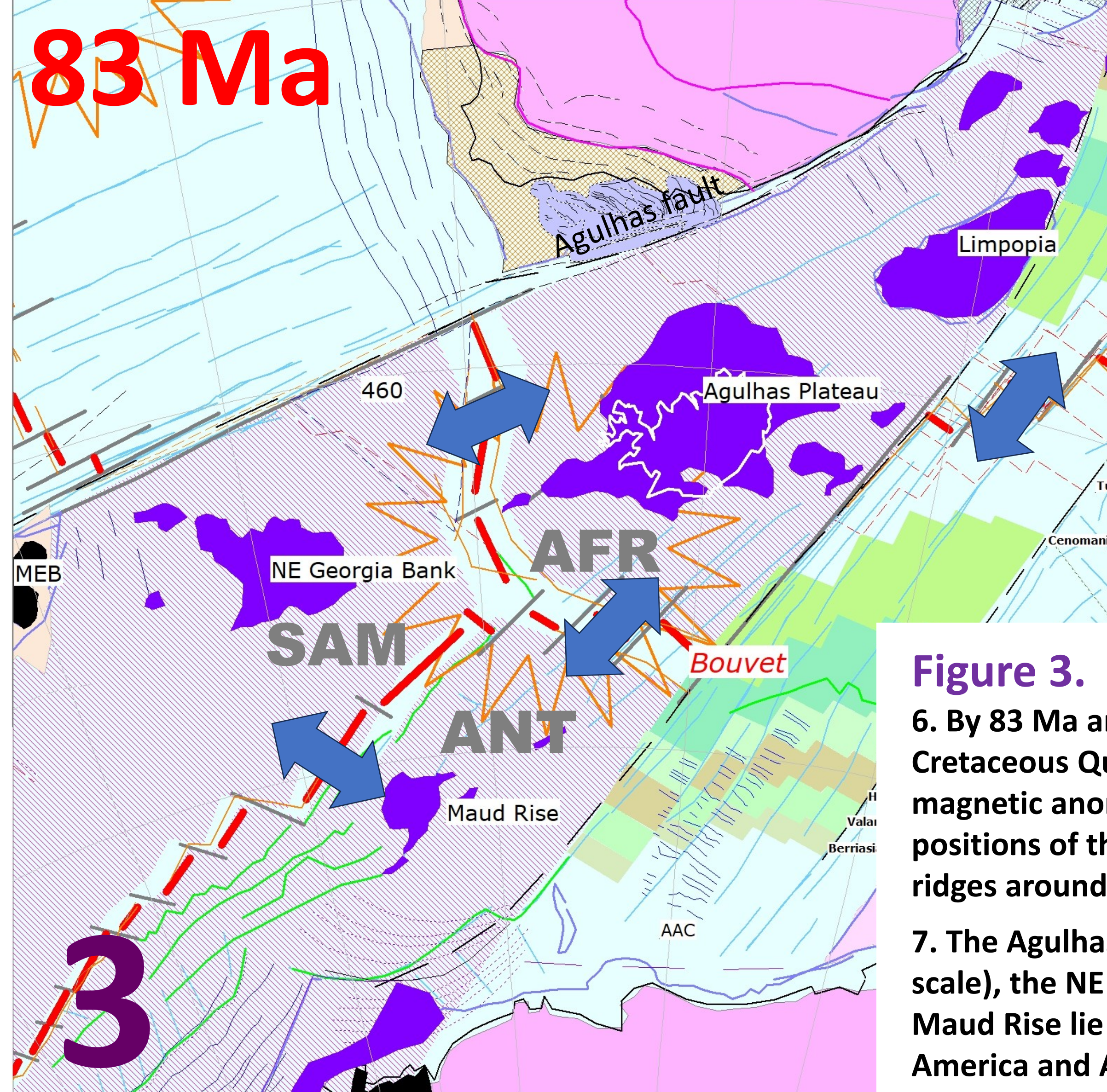


Figure 3.

6. By 83 Ma and the end of the Cretaceous Quiet Zone, marine magnetic anomalies c34 define the positions of the three mid-ocean ridges around the triple junction.
7. The Agulhas Plateau (Iceland for scale), the NE Georgia Bank and the Maud Rise lie on the Africa, South America and Antarctica plates respectively.

95.00 Ma :: CR26BAAD :: Figure 5 :: 2026 March 23

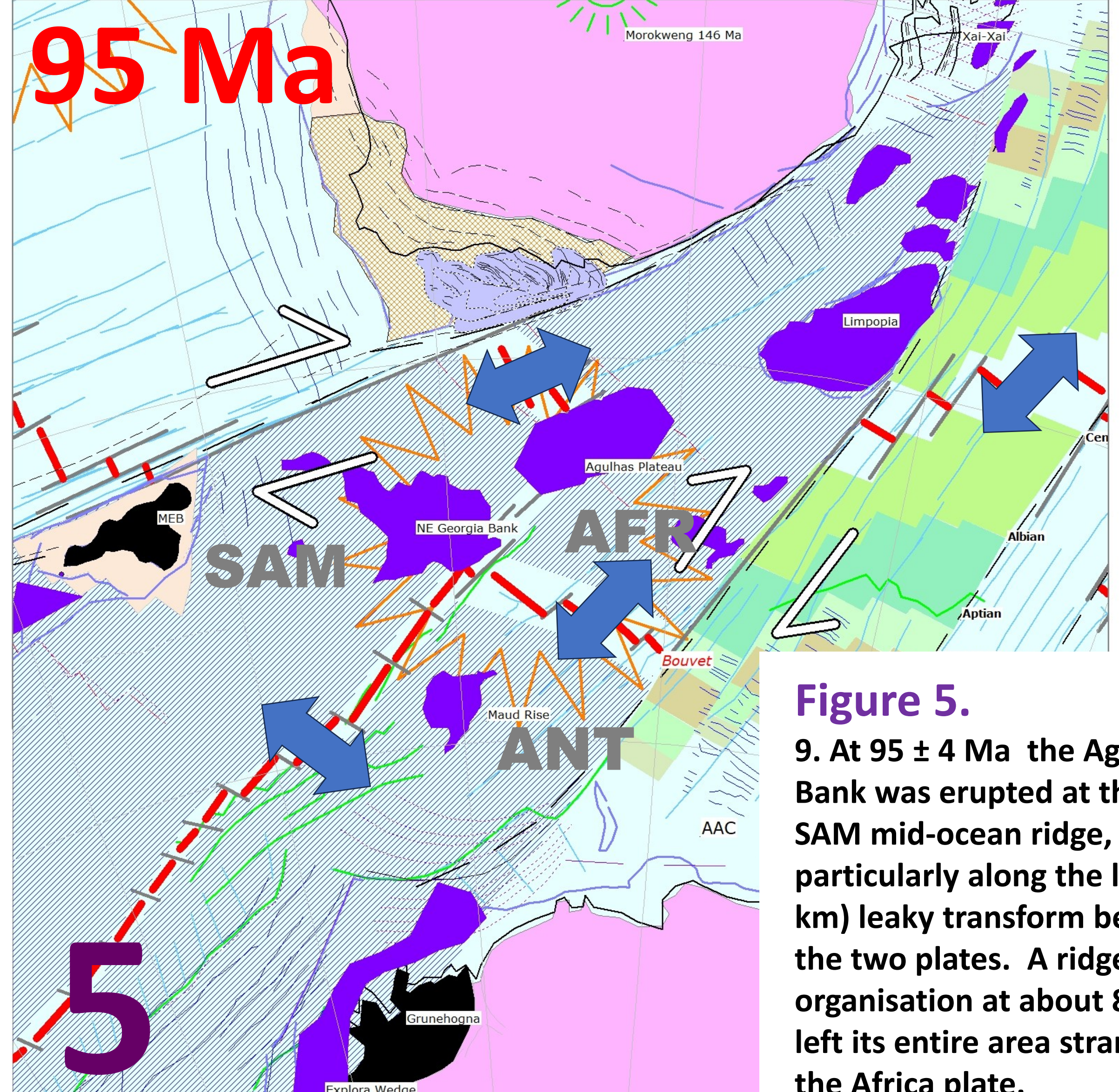


Figure 5.

9. At  $95 \pm 4$  Ma the Agulhas Bank was erupted at the AFR-SAM mid-ocean ridge, particularly along the long (500 km) leaky transform between the two plates. A ridge re-organisation at about 86 Ma left its entire area stranded on the Africa plate.

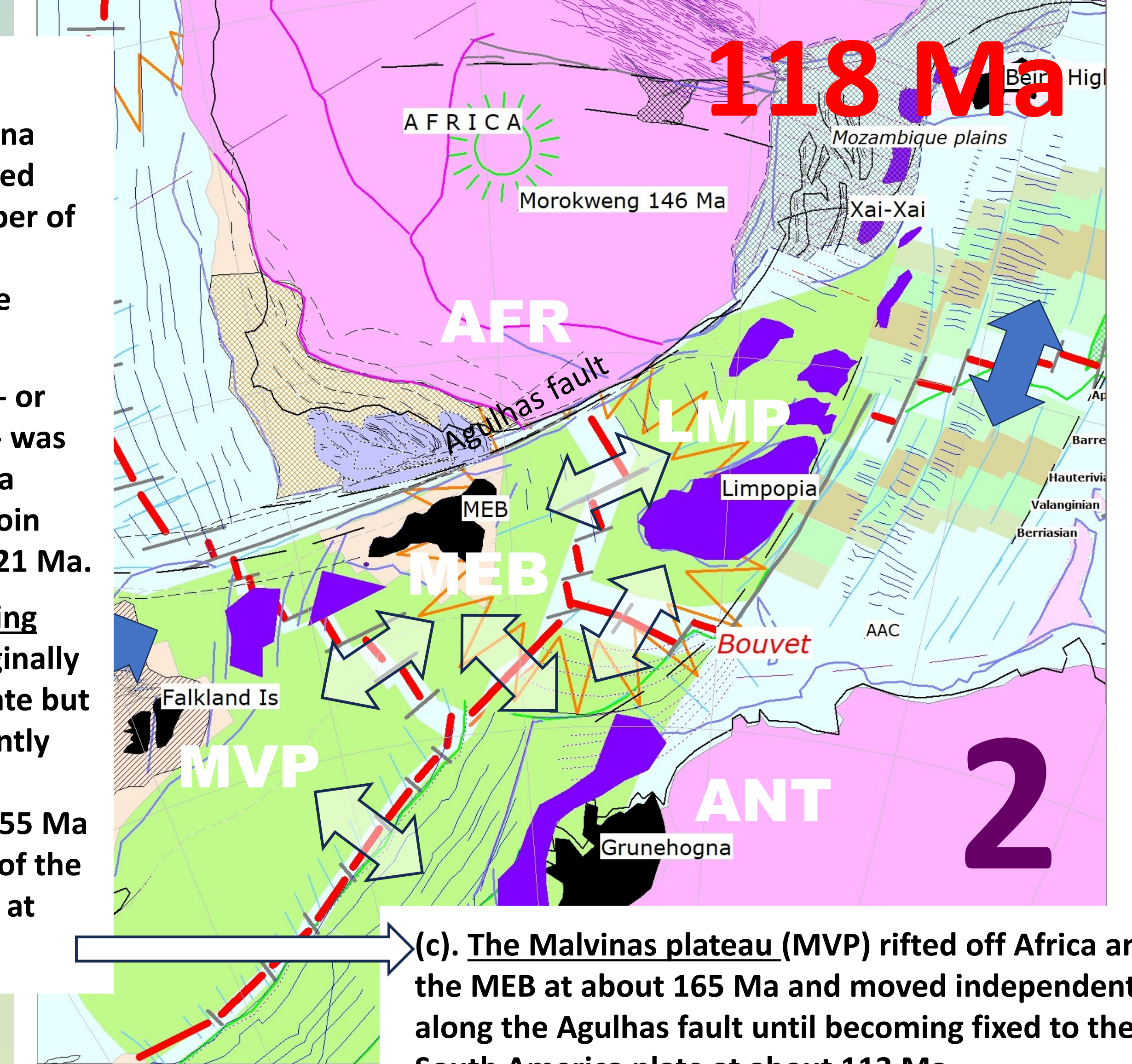
Figure 2.

4. Once the Gondwana supercontinent started fragmenting, a number of smaller continental fragments have to be considered:

(a). Limpopia (LMP) - or Mozambique Ridge - was part of the Antarctica plate, rafting off to join Africa from 142 to 121 Ma.

(b). The Maurice Ewing Bank (MEB) was originally part of the Africa plate but travelled independently westward along the Agulhas fault from 155 Ma until becoming part of the South America plate at about 113 Ma.

118.00 Ma :: CR26BAAD :: Figure 2 :: 2026 March 23



(c). The Malvinas plateau (MVP) rifted off Africa and the MEB at about 165 Ma and moved independently along the Agulhas fault until becoming fixed to the South America plate at about 113 Ma.

5. The mid-ocean ridges between all these minor conjugate margins have been considered in the model.

100.00 Ma :: CR26BAAD :: Figure 4 :: 2026 March 23

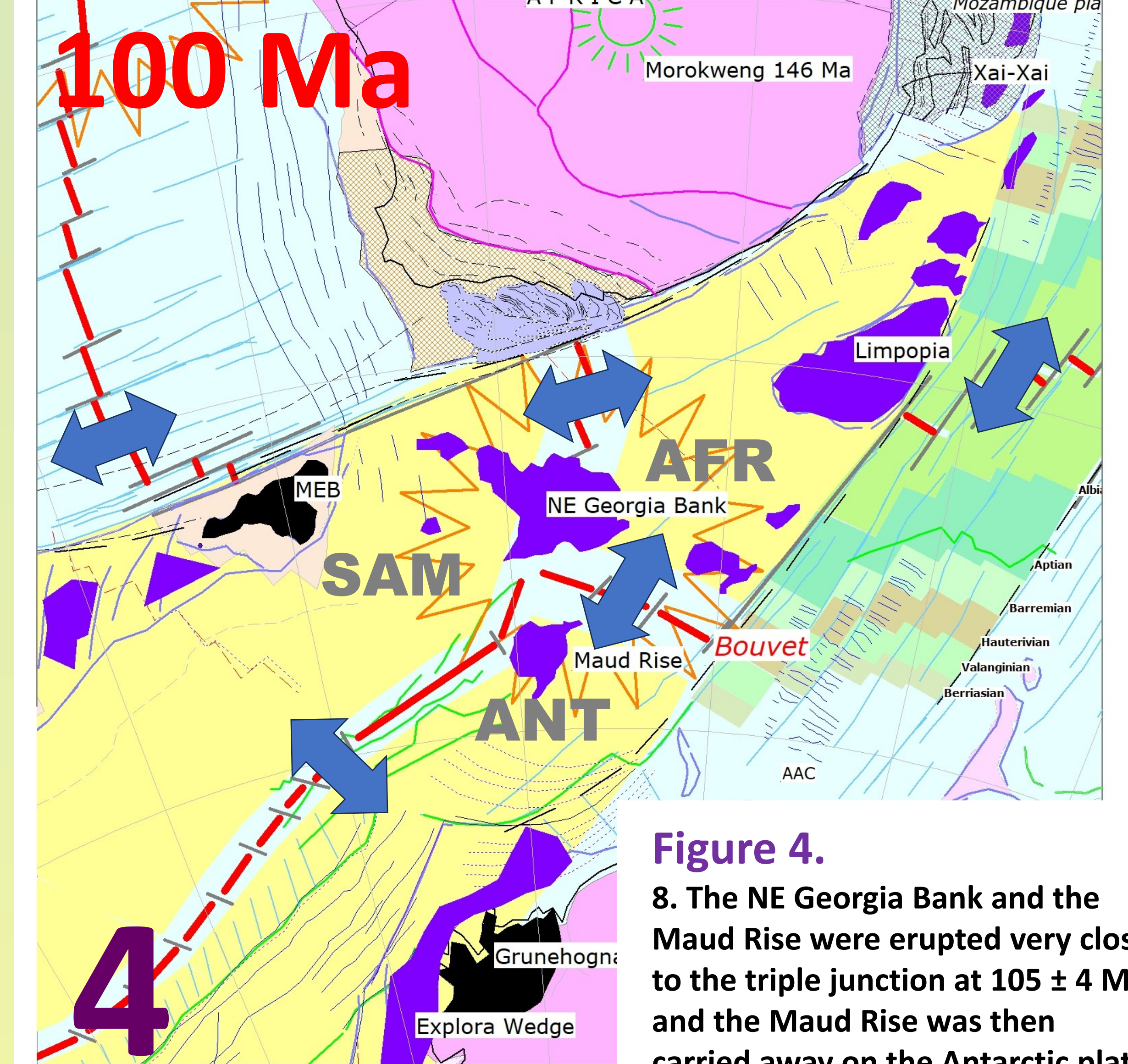


Figure 4.

8. The NE Georgia Bank and the Maud Rise were erupted very close to the triple junction at  $105 \pm 4$  Ma and the Maud Rise was then carried away on the Antarctic plate.

## Nota bene:

10. All the 'action' (including several more ridge reorganisations) takes place very close to the Bouvet plume head, shown by the orange star, 1000 km in diameter. The shorter ridges tend to die out once they become distant from it.

11. Scan the QR code to watch the whole process in animation.

Many more examples of the plate tectonics of Gondwana dispersal:

[www.reeves.nl/gondwana](http://www.reeves.nl/gondwana)

